

EE5203 L04 Pre-Lab Quiz

Pointers, masks, and the bare-metal blink - Basri Erdogan

Preparation time: 5-8 minutes **Attempts:** Two; highest score recorded **Bring to the lab:** Your answer to Question 5

```
1.  int x = 5;
    int *p = &x;
    *p = 9;
```

What is the final value of x?

- 5
 - 9
 - The address of x.
 - Undefined.
2. What does the expression `GPIOA->MODER` select?
- Bit 5 of port A.
 - The address of the GPIOA structure.
 - The `MODER` member of the register structure that the CMSIS pointer `GPIOA` points to.
 - A HAL library function.
3. Which statement clears **only** bit 7 of `value`?
- `value &= ~(1U << 7);`
 - `value |= (1U << 7);`
 - `value = ~(1U << 7);`
 - `value ^= (1U << 7);`
4. Why must `RCC->AHB1ENR |= (1U << 0);` execute before any `GPIOA` write?
- It sets PA5 high for the first time.
 - An unlocked peripheral ignores register writes.
 - It clears the `MODER` register to zero.
 - The compiler requires `RCC` before `GPIO` in the source order.
5. Complete this entry ticket before class:

Prediction	Your answer
Hex value of <code>3U << 10</code>	
Hex value of <code>1U << 21</code> , and which pin/state it commands in <code>BSRR</code>	
Which <code>MODER</code> bits belong to PA5, and the output pattern	
First check if the build is clean but LD2 never blinks	